

Wash and Dry Produce SOP

One episode washes and dries one item of produce. It starts with one dirty item in the crate, a basin of clean water with the cradle empty, the brush and the towel on their spots, and an open place in the colander and in the bowl. It ends with that item clean and dry, lying in the colander if it is a root and in the bowl if it is a fruit, the cradle empty, and every tool back on its spot. Six episodes clear one crate. One crate is one session.

The order never changes: take one item, rinse it in the basin, scrub it with the brush, dry it with the towel, then sort it by type. No item is scrubbed until it has been rinsed, and no item is sorted until it is dry. The item sits in the cradle while it is scrubbed and while it is dried, so an item is never held in the air while a tool works on it.

The gripper does all the work. It takes the item from the crate, dips it in the basin, seats it in the cradle, scrubs it with the brush, wipes it with the towel, turns it over, and lays it in the colander or in the bowl. The cradle holds the item still while the brush and the towel work on it, so the gripper is free for the tool. The cradle is the only thing that ever holds an item still, and neither the cradle nor the basin is moved during a session.

Only one thing is held at a time. Nothing is passed from one grip to another, no grip is changed while an item is off the table, and nothing is dragged or slid along the table. Every pick lifts straight up off the surface before it moves anywhere.

The counts are fixed: one item per episode, one dip of 3 seconds, one drip of 2 seconds over the basin, two scrubs, one on each side, two wipes, one on each side, and six items to a crate. There is no second dip and no extra scrub once the towel has been picked up. Water that lands outside the basin is left exactly where it falls, tagged, and the episode goes on. Nothing is ever tidied up on camera.

Setup

Finish both checklists before starting an episode.

Cell configuration

Environment camera: 900 mm.

cell_type: single_arm

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole tabletop in one shot: the crate spot, the brush spot, the basin with the cradle in it, the towel spot, the colander spot and the bowl spot.

3. The item stays in frame from the moment it leaves the crate until it lands in the colander or the bowl, and the water line and the cradle both stay in frame for the whole dip.
4. The gripper is at home and open.
5. The tabletop is dry and holds nothing but the items in the Materials checklist.
6. The gripper reaches the crate spot, the brush spot, the water in the basin, the cradle, the towel spot, the colander spot and the bowl spot without stretching to a joint limit or folding in on itself. Bring a spot forward until this is true.
7. The paths from the crate to the basin, from the brush spot to the cradle, from the towel spot to the cradle, and from the cradle to the colander and to the bowl are clear. Carry one item along each path by hand during setup: it must clear the basin wall and the container rims. Widen the gaps until it does.
8. No path from the basin to a container passes over the towel spot, so a wet item never travels over the dry towel.

Materials checklist

1. Six dirty items stand in the crate at the crate spot at the back left. Each one is a root or a fruit, each one is whole and firm, and none is bruised, split or soft.
2. Each item is small enough for the gripper to close on it squarely and hold it level, and heavy enough to stay put in the cradle while the brush works on it.
3. The basin sits in the middle of the table, half filled with clean water. The water is deep enough to cover the biggest item and low enough that a dip does not push it over the rim.
4. The basin grips the table. Push it by hand during setup: if it slides or lifts, weigh it down or clamp it. Nothing pins the basin during the episode.
5. The cradle sits in the basin at its near side, above the water line, with its open top facing up. It is deep enough to hold one item still and shallow enough that an item lifts straight out of it. Grit and water that come off an item fall from the cradle back into the basin.
6. One brush lies flat on the brush spot at the back right, bristles down, handle toward the front. Its bristles are stiff, clean and dry at the start of a session, and none of them is bent flat or missing.
7. One towel, folded flat, lies on the towel spot to the right of the basin. It is dry and clean at the start of every episode, and it is big enough to wipe the biggest item in the crate.
8. The colander stands empty at the colander spot at the front left. Washed roots go in it and nothing else.
9. The bowl stands empty at the bowl spot at the front right. Washed fruit goes in it and nothing else.
10. The colander and the bowl are clearly different in size or colour, and the same one is the colander in every session. Both sit flat, do not rock, and are wide enough for an item to go straight down into them without being pushed.

11. A clear gap of at least a gripper's width sits around the cradle on every side, so an item lifts straight up out of it and travels away without touching a basin wall.

Workspace layout

Three rows: the crate and the brush at the back, the basin with the cradle in the middle and the towel beside it, the two containers at the front. Each item travels forward from the crate to the basin, then forward again to its container.

Back row, left to right:

1. **Crate spot**, back left. The dirty items stand here. It empties as the session runs.
2. **Brush spot**, back right. The brush lies flat here between scrubs.

Middle row, left to right:

1. **Basin**, the middle of the table, straddling the midline, half filled with clean water. Everything wet happens here.
2. **Cradle**, in the basin at its near side, above the water line. It holds one item still while the brush and the towel work on it.
3. **Towel spot**, to the right of the basin, clear of the basin rim. The folded towel lies flat here between wipes.

Front row, left to right:

1. **Colander spot**, front left. The clean dry roots lie here.
2. **Bowl spot**, front right. The clean dry fruit lies here.

No carry passes over the towel, over the brush, or over the other container, and nothing travels to the far side of the table.

Vocabulary

Item: one piece of produce that needs washing. One item per episode.

Root: a hard item with skin that is scrubbed, like a potato or a carrot. Roots go in the colander.

Fruit: a firm item with skin that is eaten, like an apple or a pear. Fruit goes in the bowl.

Type: root or fruit. The type of an item is clear at a glance and never changes during a session.

Basin: the open tub of clean water in the middle of the table. It is the wash area, and it stays where it is.

Water line: the top of the water in the basin. It sits well below the basin rim.

Cradle: the open holder in the basin that keeps one item still. It is the work area, and it stays where it is.

Seated item: an item lying in the cradle, resting on its own weight, not rocking and not leaning on a basin wall.

Dip: the item lowered under the water line and held there for 3 seconds, then lifted straight out.

Drip: the item held still over the basin for 2 seconds after the dip, so the loose water falls back in.

Rinsed item: an item that has had one dip and one drip. No grit is stuck to it in a lump.

Brush: the stiff brush used on the item. It is held by its handle, bristles down, and it touches the item and nothing else, ever.

Scrub: one pass of the brush over one side of the item, in contact the whole way, from one end to the other and back, until no dirt shows on that side.

Scrubbed item: an item with no dirt showing on any side of it.

Towel: the folded dry towel used to dry the item. It is held so a broad flat side meets the item, never by a corner and never balled up. It touches the item and nothing else, ever.

Wipe: one pass of the towel over one side of the item, pressed flat on it, until no wet shine is left on that side.

Dry item: no water drops and no wet shine anywhere on the item, top, sides or bottom.

Turn: the item lifted out of the cradle, turned a half turn, and set back in the cradle, so the side that was down is now up.

Hold: a closed empty gripper pressed onto an object to keep it still. A hold never lifts, carries or moves what it holds.

Tool home: the spot a tool returns to and lies flat on before the next step starts: the brush spot, the towel spot. A tool is never left in the basin, in the cradle, in a container or in the gripper between steps.

Release point: the gripper opens over the cradle, over the colander, over the bowl, or over a tool home. Never over the basin water, never over the other container and never over open tabletop.

Stable placement: the object lies still after release, flat on its surface, and does not rock, tip, slide or lean on anything.

Spill: water or an item outside the basin, the cradle, the colander or the bowl. It is left where it falls and cleared in the reset, never on camera.

Quarter turn: the most any tool is tilted in the gripper. Nothing is ever turned over in the air except the item, and the item is only ever turned a half turn.

Home: the resting pose, clear of the table with the gripper open.

Steps

Step 1: Take one item and rinse it

Goal: one item has had one dip and one drip, and it sits in the cradle.

The basin grips the table on its own, so nothing pins it while the item goes in.

1.1 Take one item

Close on one item in the crate, square to it, so it hangs level.

Lift it straight up off the crate and carry it forward to the basin, level the whole way.

Take one item per episode. Never lift two at once, and never drag or slide an item across the table.

Check: one item hangs level in the gripper and nothing else in the crate has moved. If two items came away together, set both back in the crate and take one again. Make no more than two extra tries.

1.2 Dip it in the water

Lower the item straight down until it is under the water line, and stop.

Hold it under the water for 3 seconds.

Lift it straight up out of the water until it is clear of the water line.

Check: the whole item went under the water and stayed under for 3 seconds. One dip, and one only. A short dip is not topped up with a second dip: leave it and let review flag the episode.

1.3 Drip it and seat it in the cradle

Hold the item still over the basin for 2 seconds, so the loose water falls back in.

Lower it into the cradle, let it take its own weight, and open the gripper.

Lift clear.

Check: the item lies in the cradle, does not rock, and does not lean on a basin wall. If it rocks or leans, lift it clear and seat it again. Make no more than one correction.

Step 2: Scrub the item

Goal: no dirt shows on any side of the item, and the brush is back on its spot.

The item stays in the cradle for this whole step. It is never scrubbed in the air.

2.1 Take the brush and scrub the top

Close on the brush handle at the brush spot, lift it straight up, and carry it to the cradle, bristles down.

Set the bristles flat on the item and draw them from one end of it to the other and back, in contact the whole way.

Work across the top and around the sides until no dirt shows on them.

Check: the brush stays in contact through the pass and the item stays in the cradle. If the item pops out of the cradle, lay the brush on its spot, seat the item again, and take the brush again.

2.2 Turn the item and scrub the other side

Carry the brush back to the brush spot, lay it flat bristles down, release, and lift clear.

Close on the item in the cradle, lift it straight up, turn it a half turn, and set it back in the cradle so the side that was down is now up.

Take the brush again and scrub that side the same way, until no dirt shows on it.

Check: both sides of the item have had one scrub. Two scrubs in total, one on each side. There is no third scrub.

2.3 Park the brush

Carry the brush back to the brush spot, lay it flat bristles down, release, and lift clear.

Check: the brush lies fully on its spot, not in the basin, not in the cradle and not over the table edge. Dirt is moved by the brush and by nothing else. A bare gripper never rubs an item.

Step 3: Dry the item

Goal: no water drops and no wet shine anywhere on the item, and the towel is back on its spot.

The item stays in the cradle for this whole step. It is never wiped in the air.

3.1 Take the towel and wipe the top

Come down on the folded towel on its spot and close on it so a broad flat side hangs below the gripper.

Press that flat side onto the item and draw it from one end of the item to the other, in contact the whole way.

Work across the top and around the sides until no wet shine is left on them.

Check: the towel presents a flat face and does not hang by a corner or slip. If it does, lay it back on its spot and take it again. Make no more than two corrective takes. If it still will not sit flat, end the episode and change the towel during reset.

3.2 Turn the item and wipe the other side

Carry the towel back to the towel spot, lay it flat, release, and lift clear.

Close on the item in the cradle, lift it straight up, turn it a half turn, and set it back in the cradle so the side that was down is now up.

Take the towel again and wipe that side the same way, until no wet shine is left on it.

Check: both sides of the item have had one wipe. Two wipes in total, one on each side. If a wet patch is still in view, make one more wipe over that patch. If the item is still wet after that, end the episode and change the towel during reset.

3.3 Park the towel

Carry the towel back to the towel spot, lay it flat, release, and lift clear.

Check: the towel lies fully on its spot, not draped over the basin, a container or the table edge.

Step 4: Sort the item by type

Goal: the item lies in the colander if it is a root and in the bowl if it is a fruit, and the cradle is empty.

One item per carry. The item is dry before it leaves the cradle, so no water is carried to a container.

4.1 Choose the container

Look at the item in the cradle. A root goes to the colander at the front left. A fruit goes to the bowl at the front right.

Choose before the item is lifted. The choice is not changed once the item is off the cradle.

Check: the container matches the type of the item. If the wrong container was chosen and the item has not yet been released, carry the item on to the right one without setting it down anywhere else.

4.2 Lay it in

Close on the item in the cradle, square to it, and lift it straight up clear of the basin.

Carry it forward to the chosen container, level the whole way, clear of the basin rim and clear of the other container.

Lower it until it rests on the bottom of the container or on the items already in it, then open the gripper and lift clear.

Never open the gripper above the rim and never drop an item in.

Check: the item lies in the right container, rests still, and does not stand proud of the rim or lean on it. If it rocks or hangs over the rim, lift it clear and lay it in again. Make no more than one correction per item.

Step 5: End the episode

Check the item: clean, dry, and lying in the right container for its type.

Check the cradle: empty, in the basin, with nothing left in it.

Check the basin: water still in it, no item left under the water.

Check the tools: the brush flat on its spot, the towel flat on its spot.

Fix a failed check only with the fix written in its step. If no fix is written, or the fix would need an earlier step run again, end the episode and reset instead.

Go home with the gripper open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Leave the items in the colander and in the bowl where they are until the crate is empty, then take both away for the next stage and stand a fresh empty colander and a fresh empty bowl on their spots.
2. Top up the crate so it holds enough dirty items to finish the session, each one whole and firm, and each one clearly a root or a fruit.
3. Shake the loose grit off the brush into the bin, dry it, and lay it flat on the brush spot bristles down with the handle toward the front.
4. Lay a dry clean towel flat on the towel spot. Replace a towel that is damp, dirty or heavy with grit. A towel is never used damp.
5. Lift any grit out of the cradle and out of the basin, and top the basin up to the water line with clean water. Change the water when it is cloudy, and always between sessions.
6. Wipe any spilled water, grit or peel off the tabletop, off the basin rim and around the containers, and dry them.
7. Check that the basin still grips the table, the cradle still holds an item without rocking, the brush bristles are still stiff, and the colander and the bowl still sit flat.
8. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

Visible cue: the brush touches an item that has not been dipped, the towel touches an item that has not been scrubbed, an item goes to a container before it has been dried, two items are washed in one episode, or a step is skipped.

SOP rule broken: Steps 1 to 4, rinse, scrub, dry, sort, in that order, one item per episode.

Coaching note: rinse it, scrub it, dry it, sort it. Nothing is scrubbed before it is wet and nothing is sorted before it is dry.

Violation: Item worked on in the air

Visible cue: the brush or the towel is used on the item while the item is held in the gripper or off the cradle, instead of seated in the cradle.

SOP rule broken: Steps 2 and 3, the item stays seated in the cradle while it is scrubbed and while it is dried.

Coaching note: the cradle holds the item and the gripper handles the tool. Seat the item and let go before a tool is picked up.

Violation: Item not seated in the cradle

Visible cue: the item is left leaning on a basin wall, rocking, or half out of the cradle, or it pops out during a scrub and the scrub goes on anyway.

SOP rule broken: Step 1.3, the item lies in the cradle on its own weight and does not rock or lean.

Coaching note: let the item take its own weight before the gripper opens. If it pops out, park the tool and seat it again.

Violation: More than one item taken

Visible cue: two items come out of the crate together, or a second item is taken before the first one is in its container.

SOP rule broken: Step 1.1, one item per episode, lifted on its own.

Coaching note: one item, start to finish. Set both back and take one again.

Violation: Dip skipped or short

Visible cue: the item goes to the cradle without going under the water, part of the item never goes under the water line, or it comes out before 3 seconds.

SOP rule broken: Step 1.2, the whole item goes under the water line and stays under for 3 seconds.

Coaching note: the whole item goes under, and it stays under while the count runs.

Violation: Second dip

Visible cue: the item goes back under the water after it has been lifted clear, at any point in the episode.

SOP rule broken: Step 1.2, one dip, and one only.

Coaching note: the rinse is one dip. A second dip makes the episode unlearnable.

Violation: Scrub skipped, short, or one side missed

Visible cue: the brush never touches the item, the brush lifts part way through a pass, a side of the item is never touched by the bristles, or the item is turned and the second side is not scrubbed.

SOP rule broken: Step 2, two scrubs, one on each side, in contact the whole way, until no dirt shows.

Coaching note: both sides get one pass each. The pass that is skipped is almost always the one that was face down.

Violation: Item rubbed by the gripper

Visible cue: the gripper rubs, pinches, or picks dirt off the item, on the item or in the cradle, instead of the brush doing it.

SOP rule broken: Step 2.3, dirt is moved by the brush and by nothing else.

Coaching note: the brush does this job. A bare gripper never rubs an item.

Violation: Towel held wrong, or the item dried without it

Visible cue: the towel is held by a corner or balled up so only a narrow line of it meets the item, or the item is dried by shaking it or by wiping it with the bare gripper.

SOP rule broken: Step 3.1, the towel is held so a broad flat side meets the item.

Coaching note: a broad flat face on the item. A narrow edge smears the water instead of taking it up.

Violation: Wet item sorted

Visible cue: water drops or a wet shine are in view on the item as it leaves the cradle, or water drips off it on the way to a container.

SOP rule broken: Steps 3 and 4, the item is dry before it leaves the cradle.

Coaching note: look at the item before the lift. Dry means no shine anywhere, bottom included.

Violation: Wrong container

Visible cue: a root is laid in the bowl, a fruit is laid in the colander, or an item is laid on the table or back in the crate instead of in a container.

SOP rule broken: Step 4.1, roots to the colander and fruit to the bowl.

Coaching note: name the type before the lift. The choice is made in the cradle, not over the rim.

Violation: Item dropped into the container

Visible cue: the gripper opens while the item is still above the rim of the colander or the bowl, or the item falls the last part of the way in.

SOP rule broken: Step 4.2, lower the item until it rests, then open the gripper.

Coaching note: lower until it touches, then release. Produce bruises where it lands.

Violation: Tool not returned to its home

Visible cue: the brush is left in the basin, in the cradle or on the table, the towel is left on the basin rim, over a container or on the tabletop, or a tool is still in the gripper when the next step begins.

SOP rule broken: Steps 2.3 and 3.3, each tool returns to its home and lies flat there before the next step starts.

Coaching note: the return is part of the step, not a gap between steps. Lower until it touches, then release.

Violation: Spill cleaned up

Visible cue: after water, grit or an item lands outside the basin, the cradle or a container, the gripper or the towel is used to sweep, wipe, push or pick it up.

SOP rule broken: Steps 1 to 4, a spill is left where it falls and is cleared in the reset.

Coaching note: cleanup motions are not part of the task. The spill is data, the cleanup is noise.

Violation: Water thrown out, or the basin moved

Visible cue: the item is swirled or shaken hard enough to throw water over the basin rim, or the basin, the cradle or a container is shoved, nudged or slid out of its spot.

SOP rule broken: Steps 1.2 and 1.3, the item is lowered and lifted straight, and the basin and the cradle stay where they are.

Coaching note: straight down, hold, straight up. Nothing on this station is pushed into line.

Violation: Carry over a working spot

Visible cue: a wet item passes over the towel spot or over the dry container, the brush passes over a container, or the towel passes over the basin water.

SOP rule broken: Steps 1 to 4, no carry passes over the towel, over the brush, or over the other container.

Coaching note: route around, not across. A wet item never travels over anything that has to stay dry.

Violation: Bad pick, carry, or release

Visible cue: an object moves from one grip to another without being set down first, a grip is shifted or seated again while the object is off the table, an item or a tool slides along the table with no daylight under it, an item hangs tilted because the grip was taken off square, or a tool is rolled past a quarter turn or turned over in the air.

SOP rule broken: Steps 1 to 4, one object at a time, lifted straight up off the surface before any sideways motion, gripped square, and never turned over in the air except for the half turn of the item in the cradle.

Coaching note: set it down to change the grip, and lift before the carry. An item that hangs tilted was taken off square.

Violation: Placement pushed, or correction limit passed

Visible cue: an item, a tool or a container is shoved, nudged or slid into position instead of being lifted clear and placed again, or a failed check is retried more often than its step allows.

SOP rule broken: Steps 1 to 4, a bad placement is corrected by lifting the object clear and placing it again, and each check allows only the correction it states.

Coaching note: nothing on this station is pushed into line. If the grip or the approach is forcing repeated correction, let review flag it and move on.

Violation: Dropped, collided, or knocked over

Visible cue: under command, the gripper or a carried object strikes the basin, the cradle, a container, the crate or a tool and knocks, tips, slides or displaces it, or an object falls short of its spot, with no gripper failure, drift or controller fault in view.

SOP rule broken: Steps 1 to 4, keep each object on a clear travel path and release only after a settled placement.

Coaching note: watch the whole path, not just the target.

Violation: Work outside the camera frame

Visible cue: a dip, a scrub, a wipe, a turn or a placement happens partly or wholly outside the environment camera frame.

SOP rule broken: Steps 1 to 4, every move happens inside the camera frame.

Coaching note: the work has drifted outside the framed area. Set the spots back to their setup positions.

Violation: Wrong episode ending

Visible cue: the episode ends with an item still in the cradle or still in the water, an item unwashed, unscrubbed or wet, a tool out of its home, short of home, or the gripper not fully open.

SOP rule broken: Step 5, check the item, the cradle, the basin and the tools, then go home with the gripper open and stop recording.

Coaching note: check first. Going home is the last thing that happens before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, throw the episode away, and never use them for coaching.

Recording stopped or paused during the episode (recording system).

Camera dropped frames or lost its feed (capture system).

Hardware fault: gripper failure, drift, controller caused collision, or motor error.

A spot sits outside comfortable reach, so the move stalls at a joint limit or has to fold in on itself to work the spot. Move the spot and run the Setup checklists again.

Bad basin: one that leaks, that slides or lifts under a correct dip, or whose water line will not stay above the biggest item.

Bad cradle: one that will not hold an item without rocking, that floats, or that sits under the water line.

Bad brush: one whose bristles are bent flat, missing or so soft that a correct pass leaves dirt behind.

Bad towel: one that is damp, dirty or heavy with grit at episode start, or too small to wipe the item. Change it before the next episode.

Bad item: one that is bruised, split, soft or mouldy, so it cannot be scrubbed without breaking up.

Bad container: a colander or a bowl that will not sit flat, or that is cracked.

Crate holding too few items to finish the session, or a basin holding too little water to cover an item. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Take one item from the crate
2. Dip the item in the basin
3. Hold the item over the basin to drip
4. Seat the item in the cradle
5. Take the brush from its spot
6. Scrub one side of the item
7. Return the brush to its spot
8. Turn the item over in the cradle
9. Take the towel from its spot
10. Wipe one side of the item dry
11. Return the towel to its spot
12. Lift the dry item out of the cradle
13. Lay the item in the colander
14. Lay the item in the bowl
15. Go home and end the episode